RAM ANALYSIS — FLEXLEU 1.0

RAM Analysis

ALTrac 6413

FlexLEU 1.x

Hitachi Rail – Internal

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01 PD 01	2023-10-18	Markus Scheuer	First Edition
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1. SCOPE

1.1 IDENTIFICATION

This document provides dependability information with respect to generic FlexLEU 1.0 variants. This document is part of all technical projects ALTRAC TSS FlexLEU comprising version 1.0.0, 1.0.2 and 1.0.3 and 1.0.4.

FlexBDD, a balise driver for SEC / SEA, FlexBD, a replacement for balise driver, DCA3-CIF, a balise driver for DCA 3.0 and EXS, an optional extender slice for splitting the slice bus into segments, are not within the scope of this document.

1.2 INTENDED AUDIENCE

The intended audience for this document is:

- System Architect
- Hardware Architect
- Quality Assurance Manager
- Safety Assurance Manager
- RAM Assurance Manager
- Verifier
- Validator

1.3 DOCUMENT OVERVIEW

Chapter 1 presents the scope of the document.

Chapter 2 presents the referenced documents.

Chapter 3 provides system relevant aspects together with references to system documentation.

Chapter 4 presents principles of reliability calculations used for calculating MTBF and availability figures.

Chapter 5 lists the calculated failure rates of printed board assemblies and of the FlexLEU reference setups PV 1.x.

Chapter 6 provides MTBF and availability figures of FlexLEU variants and analyses the fulfilment of the corresponding requirements.

Chapter 7 lists and evaluates all dependability requirements which are subject for this RAM Analysis.

Chapter 8 provides a short summary.

1.4 DOCUMENT MANAGEMENT

The management of this document will be performed according to the Document Management Rules DMR and the Document List and Review Matrix [DLRM].

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1.5 CHANGE HISTORY

1.5.1 EDITION 01

- First Edition – created for FlexLEU 1.0.0, 1.0.2 and 1.0.3.

1.5.2 EDITION 02

- Document updated to reflect FlexLEU 1.0.4.
- Referenced list updated
- FlexLEU component switch added in Chapter 3.1.
- Table 7, Table 9, Table 13 and Table 15 updated to include the FlexLEU switch.
- Requirements for the FlexLEU switch added in 7.1.1 and 7.1.3 ([FlexLEU-SSS-HW-REQ-0670], [FlexLEU-SSS-HW-REQ-0671], [FlexLEU-SSS-HW-REQ-0658] and [FlexLEU-SSS-HW-REQ-0661]).
- Fulfilment of the new requirements demonstrated in chapters 7.2.1 and 7.2.3.

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

Id	Reference
[ARPA_BIS.A]	AlcaRel Reliability Prediction Report - BIS.A 3BU 83022 AAAA DUAPA; Ed. 03
[ARPA_BIS.B]	AlcaRel Reliability Prediction Report - BIS.B 3BU 83022 BAAA DUAPA; Ed. 03
[ARPA_CSS.A]	AlcaRel Reliability Prediction Report - CSS.A 3BU 83032 AAAA DUAPA; Ed. 03
[ARPA_CSS.B]	AlcaRel Reliability Prediction Report - CSS.B 3BU 83032 BAAA DUAPA; Ed. 03
[ARPA_DIS]	AlcaRel Reliability Prediction Report – DIS 3BU 83102 AAAA DUAPA; Ed. 04
[ARPA_DOS]	AlcaRel Reliability Prediction Report – DOS 3BU 83112 AAAA DUAPA; Ed. 02
[ARPA_LFC10]	AlcaRel Reliability Prediction Report - LFC10 3BU 83001 AAAA DUAPA; Ed. 01
[ARPA_PACS]	AlcaRel Reliability Prediction Report – PACS 3BU 83012 AAAA DUAPA; Ed. 03
[ARPA_PACS_GSM_R]	AlcaRel Reliability Prediction Report - PACS Extension Board GSM-R 3BU 83013 AAAA DUAPA; Ed. 01
[ARPA_PACS_LTE]	AlcaRel Reliability Prediction Report - PACS Extension Board LTE 3BU 83014 AAAA DUAPA; Ed. 01
[ARPA_SC]	AlcaRel Reliability Prediction Report – SC 3BU 83122 AAAA DUAPA; Ed. 02
[ARPA_VSS]	AlcaRel Reliability Prediction Report – VSS 3BU 83042 AAAA DUAPA; Ed. 02
[DS_PSU_24VAC]	Datasheet - Power Supply Slice 110-230 VAC / 24 VDC 36 W, PSU, FlexLEU 1.x 3BU 83081 AAAA 571EN, Ed. 02
[DS_PSU_24VDC]	Datasheet - Power Supply Slice 110-230 VAC / 24 VDC 36 W, PSU, FlexLEU 1.x 3BU 83081 CAAA 571EN, Ed. 02
[DS_SWITCH]	Datasheet - Ehternet Switch managed Type Lynx-3306-F2G-T4-LV 4-Port RJ45 / 2-Port SFP 3BU 15128 AAAA 122EN
[RAMAna_LEU_3.6]	Reliability Analysis, TSS LEU PV 3.x; 3BU 80811 3006 DUAPA; Ed. 12
[PrelimRAMAna_FlexLEU]	Preliminary RAM Analysis - FlexLEU 1.0; 3BU 80811 1540 DUAPA, Ed. 02
[RAM_HW_FMEA_FlexLEU]	RAM HW FMEA FlexLEU 1.x 3BU 80811 1533 DUAPA, Ed. 02
[SDD_SL_FlexLEU]	Sub Design Specification, Slices, FlexLEU 1.x 3BU 80811 1528 PEAPA, Ed. 03
[SDD_SYS_FlexLEU]	System Design Description, System, FlexLEU 1.x 3BU 80811 1520 PEAPA, Ed. 04
[SSS_HW_FlexLEU]	Sub-System Specification, HW, FlexLEU 1.x 3BU 80811 1530 DTAPA, Ed. 04
[SSS_SL_FlexLEU]	Sub-System Specification, Slices, FlexLEU 1.x 3BU 80811 1528 DTAPA, Ed. 05
[SSS_SW_FlexLEU]	Sub-System Specification, SW, FlexLEU 1.x 3BU 80811 1550 DTAPA, Ed. 02
[SSS_SYS_FlexLEU]	System Specification, System, FlexLEU 1.x 3BU 80811 1520 DTAPA, Ed. 05

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Id	Reference
[TR_2022-14]	RAM Statement FlexLEU 1.0 - Fulfilment of MTBF requirements

Table 1: Referenced Documents

2.2 STANDARDS AND PROCEDURES

Id	Reference
[Birolini]	Reliability Engineering. Theory and Practice; Alessandro Birolini; ISBN: 354040287X
[DLRM]	Document List and Review Matrix for engineering projects; 83530224-DOC-AUT-EN-012
[DMR]	Document Management Rules; 83530061-DOC-AUT-EN-014
[EN 50126-1]	EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process
[IEC60050(192)]	IEC 60050-192:2015; International Electrotechnical Vocabulary (IEV) - Part 192: Dependability
[MIL-HDBK217F]	US MIL-HDBK 217F, Reliability prediction for electronic equipment
[THALES_FAILRATE]	Reliability Standard Failure Rate Factor; 1AA 00210 0001 BCZZA, Ed. 04; (THALES Standard)
[Unisig_Subset_36]	FFFIS for Eurobalise; Subset 036, Version 4.0.0
[Unisig_Subset_44]	FFFIS for Eurobalise; Subset 044, Version 4.0.0.

Table 2: Standards and Procedures

2.3 TERMS, ABBREVIATIONS AND ACRONYMS

Term	Meaning
AITrac	Thales Train Control System
AC	Alternating Current
Availability	Probability that an item is in a state to perform a required function under given conditions at a given instant of time, assuming that the required external resources are provided. [IEC60050(192)] A qualitative definition, focused on *ability* rather than on *probability* is also usual. The *steady-state value* of the point availability can be expressed in general by $\frac{MTTF}{MTTF + MTTR}$. [Birolini]
BP	Backpanel
BIS	Balise Interface Slice
CSS	Current Sensor Slice
DC	Direct Current

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Term	Meaning
DCA	Distributed Control Architecture
DCA3-CIF	Distributed Control Architecture 3.0 C InterFace
Dependability	Ability to perform as and when required. Note 1 to entry: Dependability includes availability (192-01-23), reliability (192-01-24), recoverability (192-01-25), maintainability (192-01-27), and maintenance support performance (192-01-29), and, in some cases, other characteristics such as durability (192-01-21), safety and security. [IEC60050(192)]
DIS	Digital Input Slice
DOS	Digital Output Slice
ETCS	European Train Control System
Failure rate	Limit for $\delta t \to 0$ of the probability that an item will fail in the time interval $(t, t+\delta t]$, given that the item was new at $t=0$ and did not fail in the interval $(0,t]$, divided by δt. [Birolini]
Fault tree	A logic diagram showing which fault modes of subitems or external events, or combinations thereof, result in a given fault mode of the item. [IEC60050(192)]
FlexBD	Flexible Balise Driver
FlexBDD	Flexible Balise Driver Distributed
FlexLEU	Flexible Line Side Electronic Unit
FiT	Failures in time (unit: $10^{-9}h^{-1}$)
Ground Benign (GB)	Nonmobile, temperature and humidity controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos. [MIL-HDBK217F]
Ground Fixed (GF)	Moderately controlled environments such as installation in permanent racks with adequate cooling air and possible installation in unheated buildings; includes permanent installations of air traffic control radars and communication facilities. [MIL-HDBK217F]
GSM-R	Global System for Mobile communication - Railway
LIC	LEU Information Coupler
LEU	Lineside Electronic Unit
LFC10	Low Footprint Core 1.0
LMD	LEU Management and Diagnosis
LTE	Long Term Evolution
Mil-HDBK 217F	The Military Handbook is published by the US Department of Defense, and provides mathematical models for electrical, electronic and electro-mechanical components. Each model calculates the component failure rate for given combinations of environmental conditions, component value and type, electrical stress conditions and manufactured quality level
MTBF	The expectation of the operating time between failures. [IEC60050(192)] $MTBF$ is usually reserved for items with *constant* failure rate λ. In this case, $MTBF = 1/\lambda$ is the expected value (mean) of the *exponentially distributed item's failure-free operating time*. In particular $MTBF \approx T/k$, where T is the given, *fixed* cumulative operating time and k the total number of failures (failed items) during T. [Birolini]

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Term	Meaning
MTTR	The expectation of the time to restoration. [IEC60050(192)] $MTTR$ is the expected value (mean) of an item's repair time. It is obtained from the distribution function $G(t)$ of the repair time as $MTTR = \int_0^\infty (1 - G(t))dt$. For reliability or availability calculation a constant repair rate μ (i.e. exponentially distributed repair times with $\mu = 1/MTTR$) can often be used to get approximate results. [Birolini]
NFF	No Fault Found
PAB	Primary Adaptation Board
PABL	Primary Adaptation Board Low resistance
PACS	Processing And Communication Slice
PBA	Printed Board Assembly
PCB	Printed Circuit Board
PDU	Power Device Unit
PIC	Processor Interface C
PICCG	Processor 2x Interface C with CAN expansion galvanically separated
PIO	Parallel I/O Board
RAB	Relay Adaptation Board
SAB	Signal Adaptation Board
SC	Slice Controller
SEA	Sichere Elektronische Außenelementansteuerung (safe electronic outdoor element control)
SEC	Single Element Controller

Table 3: Terms, Abbreviations and Acronyms

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3. SYSTEM OVERVIEW

The track-side system of ALTRAC consists of data transmission points where data is transmitted to the trains.

At each transmission point, based on the given information detected by the signal adaptation the actual line speed, the static speed of the accessible line section and a location reference are transmitted to the On-Board System.

An LEU (Lineside Electronic Unit) is the device that transmits the signalling information to the passing train via balises or Euroloops.

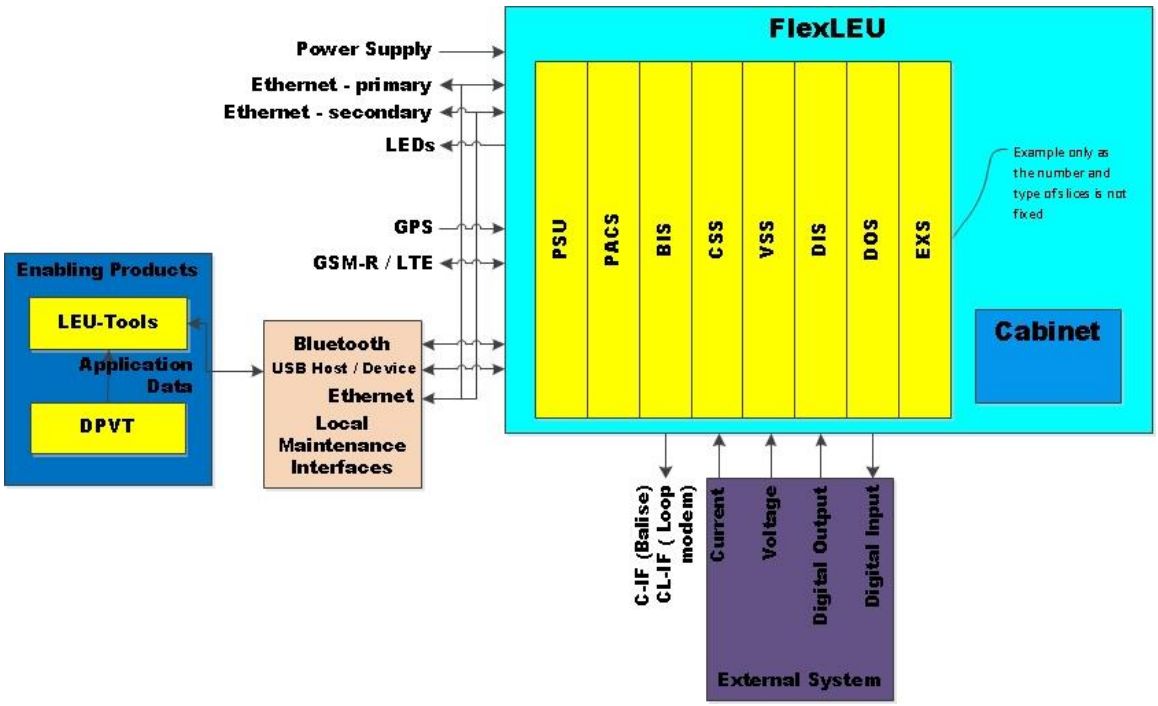
There are two different architectural characteristics:

- Stand-alone LEUs are connected directly to signals and detect the state of the signal lamps. A set of rules, included in the application data (provided by data preparation), is used to determine which ETCS telegram shall be sent via balise or loop. Stand-alone LEUs may be coupled.
- Centralized LEUs are connected to a central system (CEC). The CEC determines which telegram shall be sent to which balise (based on information from an interlocking) and sends this information to the LEUs. The LEUs pass this information to the train without further processing.

The FlexLEU is a newly designed LEU providing more flexibility compared to the legacy system. The section below presents a short overview of the FlexLEU and the components mountable within a FlexLEU rack. A detailed description can be found in the design documents [SDD_SYS_FlexLEU] and [SDD_SL_FlexLEU].

3.1 FLEXLEU COMPONENTS

The following figure taken from [SDD_SYS_FlexLEU] shows the main components of a FlexLEU and the external interfaces.



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Figure 1: FlexLEU – Overview

Figure 1 shows the decomposition of the FlexLEU into slices. The FlexLEU consists of a PSU, a PACS and other I/O slices. The cabinet is optional as either the FlexLEU designed cabinet might be used or an application specific cabinet.

All I/O slices are split into two internal units:

- generic slice controller (SC)

  The slice controller is the common communication and processing part and identical for all slices

- slice specific I/O

  These are all slice specific components that are needed to fulfill the I/O requirements of this slice.

These units are separate PBAs that are connected together and mounted in the slice box. This split allows a better re-use as the slice controller is designed only once and can be manufactured in higher quantities. The form factor of a slice is designed to fit in an ICS25 housing for mounting on a DIN rail.

Slice-Controller:
The Slice Controller is a universal converter block which acts as an interposer between the slice specific hardware and the slice bus.

HW-Slices:

- **PACS**: Functions and supervision

  Functions and supervision of the FlexLEU is concentrated at the PACS. The other slices provide a minimum functionality and are supervised by the PACS. There is never more than one PACS used inside the FlexLEU.

  The PACS is based on a carrier board where the LFC10 as processing element is located. The PACS board can be optionally extended by a GSM-R extension board resp. LTE extension board which are used for communication exchange of vital information from LIC, for integration into LMD and for maintenance activities.

- **BIS**: C-Interface/CL-Interface (2 per BIS)

  The BIS HW is an extension board for the Slice Controller. There is a variant BIS.A (3BU 83022 AAAA) delivering 2 Interfaces C for Eurobalise ([Unisig_Subset_36]). A second variant BIS.B (3BU 83022 BAAA) with one C interface and one CL1 ([Unisig_Subset_44]) interface is done via a PCB assembly variant

- **CSS**: Current input (6 per CSS).

  The CSS is a module for measuring up to six current inputs. There are two variants of the CSS slice specific hardware: A variant CSS.A (3BU 83032 AAAA) for LC (low current) for current range ± 500 mA and a variant CSS.B (3BU 83032 BAAA) for HC (high current) for current range ± 5 A.

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- **DIS**: Digital input (8 per DIS)

  The DIS HW is a module for measuring up to eight digital inputs.

- **DOS**: Digital (relay) output (4 per DOS)

  The DOS HW is a module for 4 digital outputs (2x2 outputs).

- **VSS**: Voltage input (6 per VSS)

  The VSS is a module for measuring up to six voltage inputs.

Switch:

The FlexLEU switch enables communication with other system via a fibre network and enables also a redundant communication link. It is not shown in Figure 1.

PSU:

The PSU consists of a COTS power supply supplying the slices with nominal 24V. There are two variants: One for alternating current (AC) (see [DS_PSU_24VAC]), the other for direct current (DC) (see [DS_PSU_24VDC]).

Note: Usually when denoting I/O slices by its name BIS, CSS, DIS, DOS or VSS, the slice controller (SC) is considered to be part of the I/O slice. In this RAM Analysis we *distinguish* between an I/O slice PBA and the SC PBA which is connected with the I/O slice. We consider their separate failure rates and denote in symbolic equations the functional unit I/O slice with its SC as for instance (DIS + SC), (VSS + SC), etc. Similarly, the PACS is usually considered together with its LFC10 on board. Here we also *distinguish* between the PACS PBA and the LFC10 PBA. We consider their separate failure rates and denote them in symbolic equations as (PACS + LFC10).

For historical reasons the slices within the development project are referenced using an internal definition, while the names written on the slices have been changed. The following gives the correlation:

- PSU has been renamed to FS-PSU
- PACS + LFC10 has been renamed to FS-CPU
- PACS LTE has been renamed to FS-CPU-LTE
- PACS GSM-R has been renamed to FS-CPU-GSM-R
- BIS has been renamed to FS-2C, FS-1C1CL
- CSS has been renamed to FS-6HCS, FS-6LCS
- VSS has been renamed to FS-6VS
- DIS has been renamed to FS-8DI
- DOS has been renamed to FS-4DO

3.2 LEU PV 3.6 COMPONENTS

In order to evaluate the MTBF figures of the FlexLEU components, they have to be compared with corresponding LEU PV 3.6 components. Those needed for comparison are introduced below (see [RAMAna_LEU_3.6])

- **Balise driver (BD)**

  Every LEU consists of at least 1 BD, which is equipped with 2 C-interfaces or 1 C- and 1 Loopmodem-Interface for the connection of the balises and loops.

  The following BD is only for single-BD use:

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PICC: BD with 2 C-Interfaces without CAN-Expansion board

PICL: BD with 1 C- and 1 Loopmodem-Interface without CAN-Expansion board

If two BDs are needed only a BD with a CAN-expansion board (which is necessary for communication between the two BDs) must be used. In this configuration the following type has to be installed:

PICCG2: BD with 2 C-Interfaces and a galvanically separated CAN-Expansion board.

PICLG2: BD with 1 C- and 1 Loopmodem-Interface and a galvanically separated CAN-Expansion board.

- **Primary Adaptation Board (PAB)**
  If an LEU is connected to signal lamp circuits in the primary lamp circuit, one or two PABs can be used. A PAB can handle up to 6 lamps. In the maximum configuration with 2 PABs 12 lamps can be supervised. If the LEU has only digital inputs there is no PAB which needs to be used.

- **Parallel I/O Board (PIO)**
  If an LEU is connected to digital inputs and outputs a parallel I/O board must be used. This board is able to handle up to 8 digital inputs (the external contacts must be able to switch 24V DC and currents of 10mA) and 2 digital outputs (able to switch 30V DC/250V AC at 3A with a resistance lower than 50 mΩ). If only lamp circuits have to be supervised, there is no PIO board which has to be used.

- **Power Supply Unit (PDU)**
  Every LEU has to be equipped with the appropriate power supply unit.

3.3 RAM Related Principles

RAM related principles were considered and respected during architectural design. See for instance section 3.3 in [SDD_SYS_FlexLEU] which is devoted to this topic. The following subsections can be found partly verbatim in [SDD_SL_FlexLEU]:

3.3.1 Availability and Reliability

The system is operated as single failure unit, but whenever possible, an issue of single inputs or outputs shall not cause an outage of the whole system. It has to be in case of failure, a degradation related to the specific interface only. This means for example that if one current input cannot be evaluated, the related information shall be handled as "unknown" in the subsequent evaluations. This will reduce the functionality, but allows other interfaces to stay operational.

The system is intended to be used as "last" system in the chain with local control only. Therefore, no redundancy is foreseen as the impact for the surrounding system is usually kept small.

Note: If several balises are controlled, the assignment of balises to (FlexLEU) systems has to be properly done to allow required fallback scenarios. This means for instance that one (FlexLEU) system shall be assigned to one track and another one to another track.

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The reliability targets for the different slices have been already apportioned within the system architecture design [SDD_SYS_FlexLEU]. This apportionment was also the basis for MTBF requirements of the different slices and is discussed and analysed in chapter 6, Reliability Analysis.

An operational life time of 20 years is to be achieved ([FlexLeu-SSS-REQ-0443]*)*.

3.3.2 MAINTAINABILITY

The system supports a short procedure for putting the system into operation, an effective failure detection and a short maintenance duration. This is achieved by hardware and software support together with appropriate diagnosis mechanisms.

Each slice is equipped with LED indicators. The indicators are controlled where applicable by software. Each slice has its type and a unique identifier (serial number) human readable from a label and software readable from a built-in electronic storage.

The system adheres the following maintainability principles:

- A failure of a slice shall have as small operational impact as possible
- A corrective maintenance action of a defect slice shall be possible without affecting other slices whenever possible
- Both events shall be appropriately diagnosed in order to keep life-cycle costs small.

The mean time to restore (MTTR) contains besides the repair/exchange time also the time to identify the faulty part. The identification is supported by the diagnosis functions. This can be used as basis to identify the faulty part within 30 min.

No components which require exchange within the mandatory life time shall be used (e.g. battery, fan).

3.4 BUS VS. RING ARCHITECTURE

In order to support availability and diagnostic capabilities as well as maintainability, a *ring* architecture can be used for slice communication.

The system allows the slices bus to be closed with additional connectors to a ring. This way it can conveniently handle the case of an I/O slice failure affecting the packet forwarding. This arrangement physically appears to be a ring. Here the PACS has to configure the Ethernet switch in a way that it is able to access the chain of I/O slices from both sides. The ring additionally requires that completeness of the ring is supervised. In the case the ring is broken at one point the broadcast will be performed from both sides.

This enables a better diagnosis. If for instance in a ring architecture one I/O slice is faulty, all the other I/O slices can provide their status. In a bus architecture the status of the I/O slices to the right of the faulty I/O slice is unknown. For larger FlexLEU setups with configurations containing e.g. two CSS and two BIS, there might also already be an improved operational availability.

The advantage of a ring architecture was clearly demonstrated when the RAM HW-FMEA was performed. (see [RAM_HW_FMEA_FlexLEU]). When considering the bus architecture, the table of affected I/O slices shows a triangular matrix form. In case of the ring architecture the table of affected I/O slices shows just a

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diagonal matrix form. A reasonable large setup of I/O slices has been considered, so that differences in terms of operational availability have also become visible.

The diagram from [SDD_SYS_FlexLEU] below provides an overview of the interconnectivity of the slices and their physical component exchanges with other subsystems. This slice setup serves for demonstration only as it does not consider that a FlexLEU in operation may be built from a different set of slices and the number of slices may also vary.

At the bottom row of the figure the bus connection which can be extended to a ring is shown.

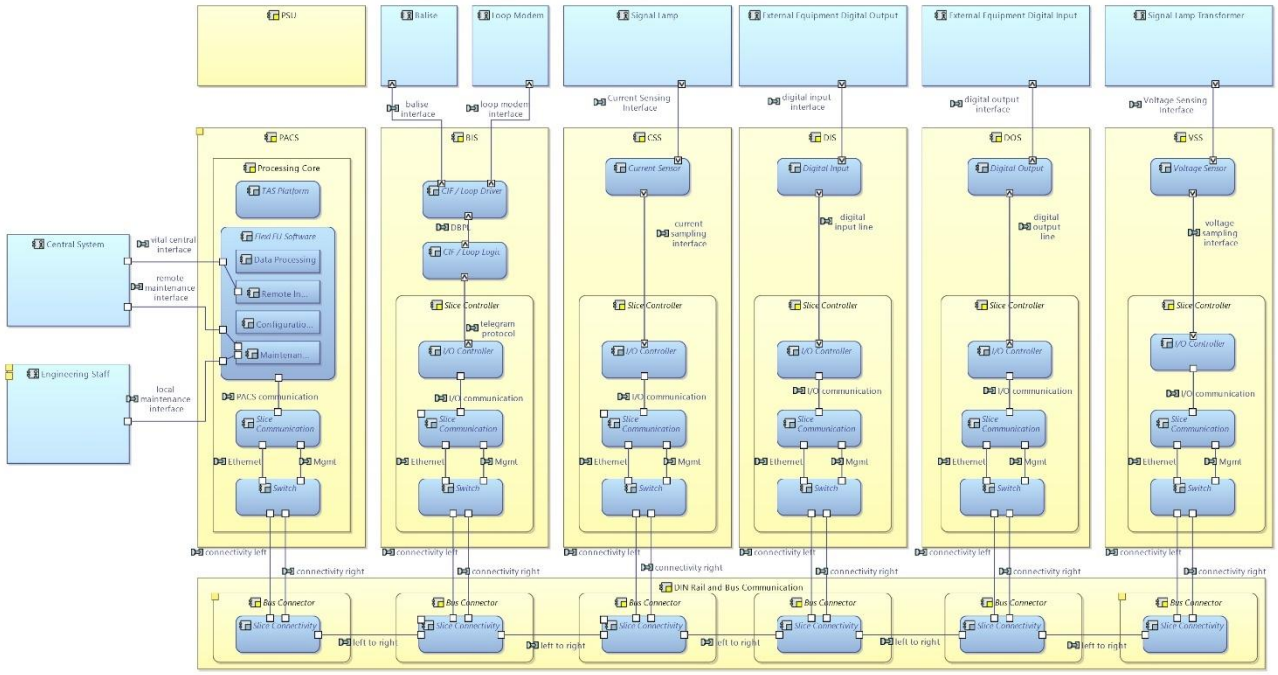


Figure 2: FlexLEU – Slices with internal and external communication

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4. RELIABILITY TECHNIQUES

4.1 LOGICAL EQUATIONS

The calculations were performed using fault tree analysis techniques and are based on so-called logical equations. The logical equations represent all possible failure combinations, which can lead to a system failure. Starting with the undesired top event all successive subordinate events are identified until the basic events are reached. The basic events are failure events that cannot be refined, without limiting further development. This means that those events are the failures of individual PBAs and devices, respectively. At this level the failure behavior of an item is derived from the constant failure rate assigned.

The logical equations use operators to combine failure events from simple components to failure events of more complex components. The operators are described in the table below. The system failure rate λ_s denotes the failure rate calculated from the sub-system failure rates λ_1, λ_2 and the repair rates μ_1, μ_2.

Meaning	Example	Failure rate λ_S
+ **denotes logical** OR	$comp_S = comp_1 + comp_2$	$\lambda_S = \lambda_1 + \lambda_2$
* **denotes logical** AND	$comp_S = comp_1 * comp_2$	$\lambda_S = \frac{\lambda_1 \lambda_2 (\mu_1 + \mu_2)}{\mu_1 \mu_2}$
$k\ comp_1 + \cdots + comp_n$ **denotes** "k out of n"	$comp_S = < k\ comp_1 + \cdots + comp_n >$	$\lambda_S \approx k\lambda \binom{n}{k} \left(\frac{\lambda}{\mu}\right)^{n-k}$

Table 4: Operators in logical equations

It is assumed that all these parts have a constant failure rate during their useful life phase. Wearing parts in the logical equations are not considered.

4.2 FAILURE RATES OF COMPONENTS

The failure rates of components were calculated by following a standard method described in [THALES_FAILRATE].

This method is principally based on the models of [MIL-HDBK217F]. In order to avoid too conservative data, a corrective factor is applied, reflecting the real failure behaviour in the field. For the track-side system a *ground benign and ground fixed* environment is assumed. Sometimes, in case of similar HW-variants, the representative with the most conservative failure rate was used.

Only randomly occurring hardware failures are considered. Faults of a systematic nature are minimised by following the rules of [EN 50126-1] during the development process.

4.3 FAILURE RATES OF PBAS AND DEVICES

The calculation of the overall failure rate of PBAs and devices is done by summing up all component failure rates located on it (parts stress method according to [MIL-HDBK217F]). For purchased devices that do not come with reliability figures, the failure rate of similar devices (previously calculated with standard methods) is taken.

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4.4 CALCULATION OF SYSTEM DEPENDABILITY QUANTITIES

Calculations of system quantities are based on the constant failure rates of PBAs and devices and the average repair time (MTTR) of 1 hour ([FlexLEU-SSS-REQ-0093]).

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5. FAILURE RATES OF FLEXLEU VARIANTS

In order to define reasonable MTBF requirements for the FlexLEU components a set of reference setups has been specified. The equivalent function of these setups in terms of LEU PV 3.6 components resulted in failure rates which were the basis for MTBF requirements.

5.1 FLEXLEU REFERENCE SETUPS

These reference setups were given in characteristic small and large and for environmental conditions indoor and outdoor (trackside). From [SSS_SYS_FlexLEU]:

- Small FlexLEU setup indoor (FlexLEU-SSS-REQ-0098)
- Small FlexLEU setup outdoor (FlexLEU-SSS-REQ-0099)
- Large FlexLEU setup indoor (FlexLEU-SSS-REQ-0100)
- Large FlexLEU setup outdoor (FlexLEU-SSS-REQ-0101)

These setups are used to characterise MTBF requirements for FlexLEU based upon comparable configurations of LEU 3.x. A cabinet which is not in the focus of the comparison of these setups is not explicitly considered in MTBF calculations, as it has typically only minor influence.

LEU PV 3.6 Setups		MTBF [y] required	
small	PICCG2K + PABL + BP + PDU		
large	2*PICCG2 + 2*PABL + PIO + BP + PDU		
FlexLEU Setups		**indoor**	**outdoor**
small	(PACS+LFC10) + BIS + (CSS or VSS) + 2*SC + PDU	**34.2**	**13.7**
large	(PACS +LFC10) + 2*BIS + CSS + VSS + DIS + DOS + 6*SC + PDU	**17.1**	**7.0**

Table 5: FlexLEU Setups

Table 5 provides logical equations for small and large FlexLEU setups. Note that the small setup lists 2 SCs as there are two I/O slices in use and each of them needs to be connected with a slice controller. Analogously there are 6 SCs given in the large setup.

The required MTBF figures are given for indoor (GB 40 °C) and outdoor (GF 40 °C) environment.

Logical equations from components of the LEU PV 3.6 setups are also stated. They are comparable in functionality and serve for comparison of their MTBF values. In terms of reliability figures the following is given:

	FiT		MTBF [h]		MTBF [y]	
	GB 40°C	**GF 40°C**	**GB 40°C**	**GF 40°C**	**GB 40°C**	**GF 40°C**
small	3 333	8 333	300 000	120 000	34.2	13.7
large	6 667	16 393	150 000	61 000	17.1	7.0

Table 6: FlexLEU Setups – Required FiT and MTBF

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In order to provide MTBF requirements for the FlexLEU HW components an apportionment of the total failure rate of the setup has been done according to following considerations:

- The PACS (plus LFC10) as master slice with high complexity is assigned 50 % of the total failure rate.
  - From these 50 % there is 70 % assigned to LFC10 as it is considered to be the most complex part of PACS.
- The remaining 50 % of the total failure rate are equally split to each of the two slices (small setup) as well as the PDU.

This breakdown was considered to be a rough, but plausible estimation. The following table shows the apportionment of the reference failure rate among the components considered in the RAM analysis. The reference failure rate is derived from the required MTBF of the small reference setup:

	FiT		MTBF [h]	
	GB 40°C	**GF 40°C**	**GB 40°C**	**GF 40°C**
PACS	1 667	4 167	600 000	240 000
BIS				
CSS				
DIS	556	1 389	1 800 000	720 000
DOS				
VSS				
PSU	556	1 389	1 800 000	720 000
Switch	556	1 111	1 800 000	900 000
Reference Failure Rate	**3 333**	**8 333**	**300 000**	**120 000**

Table 7: MTBF requirements and Failure Rate Apportionment

For PACS and LFC10 the apportionment is:

	FiT		MTBF [h]	
	GB 40°C	**GF 40°C**	**GB 40°C**	**GF 40°C**
PACS	1 667	4 167	600 000	240 000
LFC10	1 166	2 916	860 000	342 000

Table 8: Failure Rates - Apportionment PACS - LFC10

5.2 FLEXLEU COMPONENTS

The calculation of the HW-slices was performed and documented in the AlcaRel Reliability Prediction Reports [ARPA_BIS.A], [ARPA_BIS.B], [ARPA_CSS.A], [ARPA_CSS.B], [ARPA_DIS], [ARPA_LFC10], [ARPA_PACS], [ARPA_PACS_GSM_R], [ARPA_PACS_LTE] [ARPA_SC] and [ARPA_VSS].

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Table 9 below lists the calculated failure rates of the HW-slices, the failure rate of the PDU and Switch, according to the manufacturers datasheet [DS_PSU_24VAC], [DS_PSU_24VDC] and [DS_SWITCH] respectively.

Note that as stated in section 3.1, the failure rate of each slice has to be taken together with the failure rate of the slice controller (SC) in order to represent a full functioning unit.

Product FlexLEU 1.0	Item	Identification	FIT		Origin
			GB 40	GF 40	
BIS	BIS.A	3BU 83022 AAAA	1 062	2 250	ALCAREL
	BIS.B	3BU 83022 BAAA	976	2 109	
CSS	CSS.A	3BU 83032 AAAA	373	831	
	CSS.B	3BU 83032 BAAA			
DIS	DIS	3BU 83102 AAAA	972	1 849	
DOS	DOS	3BU 83112 AAAA	266	596	
PACS	PACS	3BU 83012 AAAA	672	1 458	
	LFC10	3BU 83031 AAAA	603	1 804	
	GSM-R Ext. Bd.	3BU 83013 AAAA	662	1 318	
	LTE Ext. Bd.	3BU 83014 AAAA	766	1 496	
VSS	VSS	3BU 83042 AAAA	397	829	
SC	SC	3BU 83122 AAAA	835	2 045	
Switch	Lynx-3306	3BU 15128 AAAA	339	677	Manufacturer[1]
	Lynx-3310	3BU 15128 ABAA	513	1 027	
PSU	PSU	3BU 83081 AAAA	58	257	Manufacturer[2]

Table 9: Failure Rates FlexLEU

These failure rates represent a conservative approach based on calculations of similar items using the standard method with [THALES_FAILRATE]. In order to consider the companies field experience together with this standard method in comparison with the conservative approach using [MIL-HDBK217F] some devices have been adapted.

[1] For reasons of simplicity the more conservative value is taken in further calculations

[2] For reasons of simplicity the more conservative value is taken

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5.3 LEU PV 3.6 COMPONENTS

The Table 10: Failure Rates LEU PV 3.6 below lists the calculated failure rates of the LEU PV 3.6 HW-boards and the PDU. These values will be used for comparison with those of FlexLEU PBAs. The values are taken from [RAMAna_LEU_3.6].

Product LEU PV 3.6	Item	Identification	FIT		Origin
			GB 40	GF 40	
PIC	PICCG2	3BU 80827 ABAA	1 469	3 813	ALCAREL
	PICCG2K	3BU 80893 AAAA			
PABL	PABL	3BU 80839 ACAA	1 080	2 781	
PIO	PIO	3BU 80830 AAAA	584	1 570	
		3BU 80830 ABAA			
PDU	PDU_230V	3BU 80821 AAAA	424	956	
		3BU 80821 ABAA			
	PDU_110V	3BU 80821 ACAA			
		3BU 80821 ADAA			
	PDUK_230V	3BU 80889 AAAA			
	PDU_48V DC	3BU 80841 AAAA			

Table 10: Failure Rates LEU PV 3.6

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6. RELIABILITY ANALYSIS

6.1 FLEXLEU REFERENCE SETUPS - MTBF AND AVAILABILITY

The table below shows the calculated failure rates of the FlexLEU reference setups. They are based upon the logical equations stated in Table 5 and the failure rates from Table 7.

Reference Setup	Logical Equations	FiT	
		GB 40	GF 40
small	PSU + (PACS+LFC10) + BIS + (CSS or VSS) + 2*SC	**4 393**	**10 619**
large	PSU + (PACS + LFC10) + 2*BIS + CSS + VSS + DIS + DOS +6*SC	**10 386**	**24 250**

Table 11: Reference Setups – Logical Equations

The calculation of the failure rates was done according to the description in chapter 4 and the values from Table 9.

The failure rate for the small reference setup with environmental condition GF 40 for instance is calculated as:

$$257 + (1458 + 1804) + MEDIAN(2250,2109) + MEDIAN(831,829) + 2 * 2045 \approx \mathbf{10619}$$

The table below shows the calculated MTBF and availability figures of the FlexLEU reference setups. The MTBF values (unit years) and the availability figures (unit percentage) are provided for indoor variants (environment ground benign 40 ˚C) and outdoor variants (environment ground fixed 40 ˚C). The Availability figures are calculated with an MTTR = 1h (FlexLEU-SSS-REQ-0093).

Reference Setup	FiT		MTBF [y]		Availability [%]	
	GB 40	GF 40	GB 40	GF 40	GB 40	GF 40
small	4 393	10 619	**26.0**	**10.8**	99.9996	99.9989
large	10 386	24 250	**11.0**	**4.7**	99.9990	99.9976

Table 12: FlexLEU Setups – Calculated MTBF and Availability

Note: The figures in Table 12 are the most relevant values as they correspond to the top-level requirements FlexLEU-SSS-REQ-0098 to FlexLEU-SSS-REQ-0101 (see section 7.1.1). The MTBF requirements for the different HW-slices are derived from them according to the apportionment given in Table 7 and Table 8.

6.2 FLEXLEU HW-SLICES - MTBF AND AVAILABILITY

The table below shows the calculated MTBF and availability figures of the FlexLEU HW-Slices. Here the reliability figures of the I/O slices are listed in conjunction with the slice-controller to provide the data of full functioning operative units. Also the PACS slice is shown in conjunction with the LFC10. The PACS can be optionally extended by either a GSM-R module or an LTE module. The corresponding reliability data are also stated.

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HW-Slices	FiT		MTBF [y]		Availability [%]	
	GB 40	GF 40	GB 40	GF 40	GB 40	GF 40
PACS + LFC10	1 274	3 262	89.6	35.0	99.9999	99.9997
PACS + LFC10 + GSM-R	1 936	4 580	59.0	24.9	99.9998	99.9995
PACS + LFC10 + LTE	2 040	4 758	55.9	24.0	99.9998	99.9995
BIS.A + SC	1 897	4 295	60.2	26.6	99.9998	99.9996
BIS.B + SC	1 810	4 153	63.1	27.5	99.9998	99.9996
CSS.A + SC	1 208	2 876	94.5	39.7	99.9999	99.9997
CSS.B + SC	1 208	2 876	94.5	39.7	99.9999	99.9997
DIS + SC	1 807	3 894	63.2	29.3	99.9998	99.9996
DOS + SC	1 101	2 640	103.7	43.2	99.9999	99.9997
VSS + SC	1 231	2 874	92.7	39.7	99.9999	99.9997
PSU	58	257	1 968.2	444.2	99.99999	99.99997
Switch	513	1 027	222.5	111.2	99.99995	99.99990

Table 13: FlexLEU HW-Slices – Calculated MTBF and Availability

6.3 Comparison – Required and calculated MTBF values

6.3.1 Reference Setups

A comparison of the required MTBF values and the calculated values of the reference setup is given in the table below:

Reference Setup	MTBF [y] GB 40			MTBF [y] GF 40		
	required	calculated	factor	required	calculated	factor
small	34.2	26.0	**1.3**	13.7	10.8	**1.3**
large	17.1	11.0	**1.6**	7.0	4.7	**1.5**

Table 14: FlexLEU Setups – MTBF required vs. calculated

As shown in Table 14, the calculated MTBF values are smaller than the required MTBF values. The deviation is given as proportional factor between 1.3 and 1.6.

Some reasons for this deviation are:

- New technology with extended functionality
- New modular HW-architecture (a slice controller (SC) for each I/O slice)
- Design improvements with focus on voltage stress (more components necessary)
- Design improvements with respect to diagnoses

Note: It is expected that these design improvements will help to *increase the reliability* of FlexLEU. This means that the empirical MTBF is expected to be improved, while the calculated MTBF value is smaller due to the use of additional components.

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6.3.2 HW-SLICES

A comparison of the required MTBF values and the calculated values of the HW-slices is given in the table[3] below:

HW-Slices	MTBF [y] GB 40		MTBF [y] GF 40	
	required	calculated	required	calculated
PACS + LFC10	68.5	89.6	27.4	35.0
BIS.A + SC		60.2		26.6
BIS.B + SC		63.1		27.5
CSS.A + SC		94.5		39.7
CSS.B + SC	205.3	94.5	82.2	39.7
DIS + SC		63.2		29.3
DOS + SC		103.7		43.2
VSS + SC		92.7		39.7
PSU	205.3	1 968.2	82.2	444.2
Switch	205.3	222.5	102.8	111.2

Table 15: FlexLEU HW-Slices – MTBF required vs. calculated

The MTBF requirements of the PACS with LFC10 and the PSU are fully met. The requirements of the single slices do not met the requirements. It has to be noted that these requirements were based on a rough estimation and its simple apportionment (see section 5.1).

More relevant and important is the MTBF calculation of the reference setups and their deviation of factor 1.3 to 1.6 from the top-level requirement. Some reasons for this deviation were already given in the previous section 6.3.1. A RAM analysis of this situation in terms of MTBF figures and repairs of comparable HW has been done and was summarized in the RAM statement [TR_2022-14]. The following sections in this chapter are mainly based on it.

6.4 MTBF ANALYSIS

A detailed analysis of the failure rates of the more than 50 component classes used, shows that only a few classes contribute significantly to the reduced MTBF values (see [PrelimRAMAna_FlexLEU]).

[3] There are no explicit MTBF requirements for the GSM-R and LTE extension boards.

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HW Reliability Class GF 40°C	PACS	BIS	CSS	DIS	DOS	VSS	SC	Total
DC/DC Converter	0.0	18.4	0.0	9.2	0.0	0.0	4.2	12.4
DIODE TRANSIENT SUPPR.	11.6	2.1	24.1	7.5	2.6	13.0	1.5	5.4
IC/LIN/MOS/>3K-6K	16.0	14.7	0.0	22.0	0.0	0.0	8.5	11.4
IC/LIN/MOS/0-100	0.0	4.5	23.5	1.8	0.0	23.6	0.8	5.4
OPTO COUPLER	0.0	0.0	0.0	0.0	13.4	0.0	0.0	0.4
X_TAL	0.0	0.0	0.0	0.0	0.0	0.0	15.7	3.4
Total (%)	**27.6**	**39.6**	**47.6**	**40.5**	**16.0**	**36.6**	**30.7**	**38.5**

Table 16: Failure rate percentage - Main contribution [%]

This means that possible HW changes to improve the reliability values were only promising for classes listed in Table 16. But it turned out that redesigns for significant improvements were not promising this way, which is also indicated in the next section.

6.5 REPAIR ANALYSIS

To better assess the situation, a distribution of defective HW components was analysed from the repair data of LEU. A total of approx. 1200 PIC repairs (in different variants) from nearly 6 years from all countries were analysed. The data were taken from the repair database from which a monthly snapshot is provided to the RAM-AM.

PIC repairs - All countries 2017/01 - 2022/10	cnt	[%]
Replacement	105	8.7
NFF	336	27.8
Repairs	**719**	**59.5**
open	49	4.1
Total	**1 209**	**100.0**

Table 17: PIC HW-Boards - Repair Data

As shown in Table 17: PIC HW-Boards - Repair Data, a total of 1209 PIC variants sent to the repair centre was analysed for defective components. Besides roughly 25 % NFF (no fault found) and 9 % replaced HW-boards, about 60 % of the received items were repaired. These 719 repaired items were subject of detailed statistics.

Note: About 9 % of the PICs were replaced. Some of them were diagnosed with burn damage or a defect due to overvoltage. It is expected that the design improvements of the FlexLEU will reduce this number of defects due to the improved voltage stress capability.

It turned out, that besides the class IC/LIN/MOS the reliability classes of defect components did not show a strong relationship to those from Table 16. So, the repairs did not show a clear indication for specific HW-improvement. Some more details regarding this topic are stated in [TR_2022-14].

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6.6 EMPIRICAL VS. CALCULATED MTBF

In context of FlexLEU there is a situation similar as some years ago when SEC-Signal CH was developed and the MTBF requirements were not met by the calculated MTBF figures. This led to a modification of the MTBF requirements which then could differentiate between 10 years calculated MTBF vs. 15 years empirical MTBF.

The arguments making such an approach reasonable and admissible were

- The calculated MTBF figures are typically conservative values in contrast to empirical MTBF values.
- The installed LEU in CH with roughly 1000 PIC (663 PICC, 270 PICL) at that time and some million hours total in operation provided an empirical MTBF better by a factor 1.7 to 1.8 than the calculated MTBF.

Here there is a comparable situation which makes an approach of this kind reasonable. Using the factor 1.7, as was done when the new architecture SEC-Signal CH was introduced, results in MTBF values that meet the MTBF requirements.

Reference Setup	MTBF [y] GB 40			MTBF [y] GF 40		
	required	calculated	factor	required	calculated	factor
small	34.2	44.2	**0.8**	13.7	18.3	**0.7**
large	17.1	18.7	**0.9**	7.0	8.0	**0.9**

Table 18: FlexLEU Setups – MTBF with empirical factor 1.7

Table 18 has the same structure as Table 14. In column "calculated", the values are taken from Table 14 but multiplied by a factor 1.7 reflecting the empirical failure rates from field.

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7. DEPENDABILITY REQUIREMENTS FLEXLEU

7.1 DEPENDABILITY REQUIREMENTS - OVERVIEW

The following three sections named Reliability, Availability and Maintainability list the dependability requirements specified for FlexLEU. These requirements are given at system level, refined at subsystem HW slice level and more or less verbatim taken at HW component level. They are specified in [SSS_SYS_FlexLEU], [SSS_SL_FlexLEU] and [SSS_HW_FlexLEU]. The requirements listed here are given in terms of ID, type and synopsis. A description of these requirements together with an analysis is presented in the following chapter.

There are further dependability requirements which are realized at SW level (identified as [FlexLEU-SSS-SW-REQ-nnnn]). They address *availability aspects* in the context of ring architecture versus bus architecture (see for instance [FlexLEU-SSS-SW-REQ-0199] and also [FlexLEU-SSS-SW-REQ-0225]. On the other hand, they address *diagnostic capabilities*. Those requirements are not listed as the focus here is HW specific system behavior and the fulfilment of these requirements is not subject of this RAM-Analysis. Nonetheless it might be useful to point out that careful design has also been taken on diagnosis level (see for instance [FlexLEU-SSS-SW-REQ-0048], [FlexLEU-SSS-SW-REQ-0104], [FlexLEU-SSS-SW-REQ-0108] and [FlexLEU-SSS-SW-REQ-0278] in [SSS_SW_FlexLEU]).

7.1.1 RELIABILITY

Identification	Type	Requirement Synopsis
[FlexLEU-SSS-REQ-0098]	Dependability	Mean time between failures (MTBF) - Small setup indoor
[FlexLEU-SSS-REQ-0099]	Dependability	Mean time between failures (MTBF) - Small setup outdoor
[FlexLEU-SSS-REQ-0100]	Dependability	Mean time between failures (MTBF) - Large setup indoor
[FlexLEU-SSS-REQ-0101]	Dependability	Mean time between failures (MTBF) - Large setup outdoor
[FlexLEU-SSS-REQ-0443]	Constraints	Operational lifetime
[FlexLEU-SSS-SL-REQ-0527]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - PACS
[FlexLEU-SSS-SL-REQ-0528]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - PACS
[FlexLEU-SSS-SL-REQ-0725]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - BIS
[FlexLEU-SSS-SL-REQ-0726]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - BIS
[FlexLEU-SSS-SL-REQ-0807]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - CSS
[FlexLEU-SSS-SL-REQ-0808]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - CSS

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Identification	Type	Requirement Synopsis
[FlexLEU-SSS-SL-REQ-0889]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - VSS
[FlexLEU-SSS-SL-REQ-0890]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - VSS
[FlexLEU-SSS-SL-REQ-0973]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - DIS
[FlexLEU-SSS-SL-REQ-0974]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - DIS
[FlexLEU-SSS-SL-REQ-1050]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - DOS
[FlexLEU-SSS-SL-REQ-1051]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - DOS
[FlexLEU-SSS-SL-REQ-1169]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Indoor - PSU
[FlexLEU-SSS-SL-REQ-1170]	Dependability	FlexLEU slices - Mean time between failures (MTBF) - Outdoor - PSU
[FlexLEU-SSS-SL-REQ-1580]	Constraints	Design, Development and Production Requirements - Operational lifetime
[FlexLEU-SSS-HW-REQ-0053]	Dependability	PACS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0054]	Dependability	PACS HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0102]	Dependability	BIS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0103]	Dependability	BIS HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0140]	Dependability	CSS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0141]	Dependability	CSS HW - Mean time between failures (MTBF) - Oudoor
[FlexLEU-SSS-HW-REQ-0176]	Dependability	VSS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0177]	Dependability	VSS HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0211]	Dependability	DIS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0212]	Dependability	DIS HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0248]	Dependability	DOS HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0249]	Dependability	DOS HW - Mean time between failures (MTBF) - Outdoor

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Identification	Type	Requirement Synopsis
[FlexLEU-SSS-HW-REQ-0318]	Dependability	PSU HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0319]	Dependability	PSU HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0670]	Constraints	Switch HW - Mean time between failures (MTBF) - Outdoor
[FlexLEU-SSS-HW-REQ-0671]	Constraints	Switch HW - Mean time between failures (MTBF) - Indoor
[FlexLEU-SSS-HW-REQ-0519]	Constraints	PACS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0530]	Constraints	BIS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0531]	Constraints	CSS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0532]	Constraints	VSS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0533]	Constraints	DIS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0534]	Constraints	DOS HW - Operational life time
[FlexLEU-SSS-HW-REQ-0535]	Constraints	Slice Controller HW - Operational life time
[FlexLEU-SSS-HW-REQ-0536]	Constraints	PSU HW - Operational life time
[FlexLEU-SSS-HW-REQ-0537]	Constraints	Cabinet - Operational life time
[FlexLEU-SSS-HW-REQ-0658]	Constraints	Switch HW - Operational life time

Table 19: Reliability requirements – System, Subsystem, HW-Slice

7.1.2 AVAILABILITY

Identification	Type	Requirement Synopsis
[FlexLEU-SSS-REQ-0092]	Dependability	Mean time to Restore (MTTR_1)
[FlexLEU-SSS-REQ-0093]	Dependability	Mean time to Restore (MTTR_2)
[FlexLEU-SSS-REQ-0433]	Availability	FlexLEU - I/O failure behavior, overall operation
[FlexLEU-SSS-REQ-0435]	Availability	FlexLEU - I/O slice failure behavior
[FlexLEU-SSS-REQ-0436]	Availability	FlexLEU - I/O slice communication availability
[FlexLEU-SSS-SL-REQ-0531]	Dependability	Availability - FlexLEU - Mean time to Restore (MTTR_1)
[FlexLEU-SSS-SL-REQ-0532]	Dependability	Availability - FlexLEU - Mean time to Restore (MTTR_2)

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Identification	Type	Requirement Synopsis
[FlexLEU-SSS-SL-REQ-1575]	Functional Dependability	FlexLEU slices - I/O failure behvaiour, other slice operation
[FlexLEU-SSS-SL-REQ-1577]	Functional, Dependability	FlexLEU slices - I/O slice failure behaviour
[FlexLEU-SSS-SL-REQ-1578]	Functional, Dependability	FlexLEU slices - I/O slice communication availability

Table 20: Availability requirements – System, Subsystem, HW-Slice

7.1.3 MAINTAINABILITY

Identification	Type	Requirement Synopsis
[FlexLEU-SSS-REQ-0130]	Dependability	Preventive maintenance
[FlexLEU-SSS-SL-REQ-0533]	Dependability	FlexLEU slices - Preventive maintenance - PACS
[FlexLEU-SSS-SL-REQ-0731]	Dependability	FlexLEU slices - Preventive maintenance - BIS
[FlexLEU-SSS-SL-REQ-0813]	Dependability	FlexLEU slices - Preventive maintenance - CSS
[FlexLEU-SSS-SL-REQ-0895]	Dependability	FlexLEU slices - Preventive maintenance - VSS
[FlexLEU-SSS-SL-REQ-0977]	Dependability	FlexLEU slices - Preventive maintenance - DIS
[FlexLEU-SSS-SL-REQ-1054]	Dependability	FlexLEU slices - Preventive maintenance - DOS
[FlexLEU-SSS-SL-REQ-1175]	Dependability	FlexLEU slices - Preventive maintenance - PSU
[FlexLEU-SSS-HW-REQ-0055]	Dependability	PACS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0104]	Dependability	BIS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0142]	Dependability	CSS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0178]	Dependability	VSS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0213]	Dependability	DIS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0250]	Dependability	DOS HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0320]	Dependability	PSU HW - Preventive maintenance
[FlexLEU-SSS-HW-REQ-0661]	Constraints	Switch HW - Preventive maintenance

Table 21: Maintainability requirements – System, Subsystem, HW-Slice

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7.2 FULFILLMENT OF DEPENDABILITY REQUIREMENTS FLEXLEU

In this section the fulfillment of dependability requirements is analysed. Results of requirements are of one of following types:

- Requirement fulfilled
- Requirement partially fulfilled: These addresses requirements, which are only fulfilled if the empirical factor 1.7 is considered as described in section 6.6.
- Requirement not fulfilled

7.2.1 RELIABILITY

[FlexLEU-SSS-REQ-0098]: Mean time between failures (MTBF) - Small setup indoor

Description: The MTBF for a small indoor system that consists of a power supply, 2 interface C and 6 signal interface inputs shall be at least 300 000 h resp. 34.2 years.
Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The required value is MTBF = 34.2 years (see Table 14). The calculated value stated there is MTBF = 26.0 years. This is a deviation by a factor 1.3. Some reasons for this deviation are given in section 6.3.1. Adapting the MTBF value by an empirically factor 1.7 (see Table 18) which seems to be reasonable as stated in section 6.6, the requirement is fulfilled. This requirement can therefore be considered to be partially fulfilled.

Result: Requirement partially fulfilled

[FlexLEU-SSS-REQ-0099]: Mean time between failures (MTBF) - Small setup outdoor
Description: The MTBF for a small standalone outdoor system that consists of a power supply, 2 interface C and 6 signal interface inputs shall be at least 120 000 h resp. 13.7 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The required value is MTBF = 13.7 years (see Table 14). The calculated value stated there is MTBF = 10.8 years. This is a deviation by a factor 1.3. Some reasons for this deviation are given in section 6.3.1. Adapting the MTBF value by an empirically factor 1.7 (see Table 18) which seems to be reasonable as stated in section 6.6, the requirement is fulfilled. This requirement can therefore be considered to be partially fulfilled.

Result: Requirement partially fulfilled

[FlexLEU-SSS-REQ-0100]: Mean time between failures (MTBF) - Large setup indoor
Description: The MTBF for a large indoor system that consists of a power supply, 4 interfaces C, 12 signal interface inputs, 2 digital outputs and 8 digital inputs shall be at least 150 000 h resp. 17.1 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The required value is MTBF = 17.1 years (see Table 14). The calculated value stated there is MTBF = 11.8 years. This is a deviation by a factor 1.6. Some reasons for this deviation are given in section 6.3.1. Adapting the MTBF value by an empirically factor 1.7 (see Table 18) which seems to be reasonable as stated in section 6.6, the requirement is fulfilled. This requirement can therefore be considered to be partially fulfilled.

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Result: Requirement partially fulfilled

[FlexLEU-SSS-REQ-0101]: Mean time between failures (MTBF) - Large setup outdoor
Description: The MTBF for a large standalone outdoor system that consists of a power supply, 4 interfaces C, 12 signal interface inputs, 2 digital outputs and 8 digital inputs shall be at least 61 000 h resp. 7.0 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The required value is MTBF = 7.0 years (see Table 14). The calculated value stated there is MTBF = 4.7 years. This is a deviation by a factor 1.5. Some reasons for this deviation are given in section 6.3.1. Adapting the MTBF value by an empirically factor 1.7 (see Table 18) which seems to be reasonable as stated in section 6.6, the requirement is fulfilled. This requirement can therefore be considered to be partially fulfilled.

Result: Requirement partially fulfilled

[FlexLEU-SSS-REQ-0443]: Operational lifetime
[FlexLEU-SSS-SL-REQ-1580]: Design, Development and Production Requirements - Operational lifetime

Description: An operational life time of 20 years shall be assumed.

Analysis: A carefully selection of high quality HW components assures long living operation of these parts. The GTS product strategy guarantees proper replacement and availability of HW components with the required functionality for the necessary time frame, so that the operational lifetime of 20 years is guaranteed.

Result: Requirement fulfilled

[FlexLEU-SSS-HW-REQ-0519]: PACS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0530]: BIS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0531]: CSS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0532]: VSS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0533]: DIS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0534]: DOS HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0535]: Slice Controller HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0536]: PSU HW - Operational lifetime
[FlexLEU-SSS-HW-REQ-0537]: Cabinet - Operational lifetime
[FlexLEU-SSS-HW-REQ-0658]: Switch HW - Operational lifetime

Description: An operational life time of 20 years shall be assumed.

Analysis: A carefully selection of high quality HW components assures long living operation of these parts. The Hitachi product strategy guarantees proper replacement and availability of HW components with the required functionality for the necessary time frame, so that the operational lifetime of 20 years is guaranteed.

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0527]: FlexLEU slices - Mean time between failures (MTBF) - Indoor - PACS
[FlexLEU-SSS-HW-REQ-0053]: FlexLEU slices - Mean time between failures (MTBF) - Indoor
Description: The MTBF for the PACS shall be at least 600 000 h resp. 68.4 years for the variant without GSM-R or LTE modem. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

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Analysis: The calculated MTBF for the PACS (+ LFC10) for indoor environment is 89.6 years (see Table 13) which meets the requirement.

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0528]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – PACS
[FlexLEU-SSS-HW-REQ-0054] PACS HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the PACS shall be at least 240 000 h resp. 27.4 years for the variant without GSM-R or LTE modem. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the PACS (+ LFC10) for outdoor environment is 35.0 years (see Table 13) which meets the requirement.

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0725]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – BIS
[FlexLEU-SSS-HW-REQ-0102]: FlexLEU slices - Mean time between failures (MTBF) - Indoor - BIS
Description: The MTBF for the BIS shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the BIS.A (+ SC) for indoor environment is 60.2 years (see Table 13) and for the BIS.B (+ SC) for indoor environment is 63.1 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0098] and [FlexLEU-SSS-REQ-0099] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0726]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – BIS
[FlexLEU-SSS-HW-REQ-0103]: BIS HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the BIS shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the BIS.A (+ SC) for outdoor environment is 26.6 years (see Table 13) and for the BIS.B (+ SC) for indoor environment is 27.5 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0100] and [FlexLEU-SSS-REQ-0101] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0807]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – CSS
[FlexLEU-SSS-HW-REQ-0140]: CSS HW - Mean time between failures (MTBF) - Indoor
Description: The MTBF for the CSS shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

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Analysis: The calculated MTBF for the CSS.A (+ SC) and CSS.B for indoor environment is 94.5 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0098] and [FlexLEU-SSS-REQ-0099] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0808]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – CSS
[FlexLEU-SSS-HW-REQ-0141]: CSS HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the CSS shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the CSS.A (+ SC) and CSS.B for outdoor environment is 39.7 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0100] and [FlexLEU-SSS-REQ-0101] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0889]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – VSS
[FlexLEU-SSS-HW-REQ-0176]: VSS HW - Mean time between failures (MTBF) - Indoor
Description: The MTBF for the VSS shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the VSS (+ SC) for indoor environment is 92.7 years (see Table 13.) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0098] and [FlexLEU-SSS-REQ-0099] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0890]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – VSS
[FlexLEU-SSS-HW-REQ-0177]: VSS HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the VSS shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the VSS (+ SC) for outdoor environment is 39.7 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0100] and [FlexLEU-SSS-REQ-0101] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

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[FlexLEU-SSS-SL-REQ-0973]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – DIS
[FlexLEU-SSS-HW-REQ-0211]: DIS HW - Mean time between failures (MTBF) - Indoor

Description: The MTBF for the DIS shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the DIS (+ SC) for indoor environment is 63.2 years (see Table 13: FlexLEU HW-Slices – Calculated MTBF and Availability) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0098] and [FlexLEU-SSS-REQ-0099] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-0974]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – DIS
[FlexLEU-SSS-HW-REQ-0212]: DIS HW - Mean time between failures (MTBF) - Outdoor

Description: The MTBF for the DIS shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the DIS (+ SC) for outdoor environment is 29.3 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0100] and [FlexLEU-SSS-REQ-0101] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-1050]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – DOS
[FlexLEU-SSS-HW-REQ-0248]: DOS HW - Mean time between failures (MTBF) - Indoor

Description: The MTBF for the DOS shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the DOS (+ SC) for indoor environment is 103.7 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0098] and [FlexLEU-SSS-REQ-0099] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-1051]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – DOS
[FlexLEU-SSS-HW-REQ-0249]: DOS HW - Mean time between failures (MTBF) - Outdoor

Description: The MTBF for a DOS shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the DOS (+ SC) for outdoor environment is 43.2 years (see Table 13) which does not meet the requirement. It has to be noted that this requirement is just based on a simple apportionment to provide some information about the wanted order of magnitude. This requirement is subordinate to the top-level requirements [FlexLEU-SSS-REQ-0100] and [FlexLEU-SSS-REQ-0101] for the small and large indoor reference setups which are fulfilled when using the empirical weight factor.

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Result: Requirement not fulfilled

[FlexLEU-SSS-SL-REQ-1169]: FlexLEU slices - Mean time between failures (MTBF) - Indoor – PSU
[FlexLEU-SSS-HW-REQ-0318]: PSU HW - Mean time between failures (MTBF) - Indoor
Description: The MTBF for the PSU shall be at least 1 800 000 h resp. 205.4 years. Environmental condition: Ground benign 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the PSU for indoor environment is 1968.2 years (see Table 13) which meets the requirement.

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-1170]: FlexLEU slices - Mean time between failures (MTBF) - Outdoor – PSU
[FlexLEU-SSS-HW-REQ-0319]: PSU HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the PSU shall be at least 720 000 h resp. 82.2 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according ALCAREL.

Analysis: The calculated MTBF for the PSU for indoor environment is 442.2 years (see Table 13) which meets the requirement.

Result: Requirement fulfilled

[FlexLEU-SSS-HW-REQ-0670]: Switch HW - Mean time between failures (MTBF) - Outdoor
Description: The MTBF for the FlexLEU Switch shall be at least 900 000 h resp. 102.7 years. Environmental condition: Ground fixed 40 °C assumed. MTBF according to ALCAREL.

Analysis: According to the manufacturer's datasheet, the switch has the following MTBF values:
Lynx-3306-(E-)F2G-T4-LV – 1 477 000 h resp. 168.6 years.
Lynx-3310-(E-)F2G-T8-LV – 974 000 h resp. 111.2 years.

Result: Requirement fulfilled

[FlexLEU-SSS-HW-REQ-0671]: Switch HW - Mean time between failures (MTBF) - Indoor
Description: The MTBF for the FlexLEU Switch shall be at least 1 800 000 h resp. 205.5 years. Environmental condition: Ground benign 40 °C assumed. MTBF according to ALCAREL.

Analysis: According to the manufacturer's datasheet, the switch has the following MTBF values:
Lynx-3306-(E-)F2G-T4-LV – 2 954 000 h resp. 337.2 years. (see Table 13)
Lynx-3310-(E-)F2G-T8-LV – 1 948 000 h resp. 222.5 years. (see Table 13)

Result: Requirement fulfilled

7.2.2 AVAILABILITY

[FlexLEU-SSS-REQ-0092]: Mean time to Restore (MTTR_1)
[FlexLEU-SSS-SL-REQ-0531]: Availability - FlexLEU - Mean time to Restore (MTTR_1)

Description: The MTTR part consisting of repair/exchange time of a faulty system component from diagnosing the faulty component till readiness for operation shall be less than 30 minutes.

Analysis: According to the proper design of the FlexLEU, the HW modules can be exchanged easily. The replacement of a single HW module can be done within less than 30 minutes (MTTR = 1h) assuming

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that no other faults are to repair. The remaining 30 minutes of the total MTTR allowance are dedicated to fault identification.

Result: Requirement fulfilled

[FlexLEU-SSS-REQ-0093]: Mean time to Restore (MTTR_2)
[FlexLEU-SSS-SL-REQ-0532]: Availability - FlexLEU - Mean time to Restore (MTTR_2)

Description: The MTTR part consisting of diagnosis, repair/exchange and test till readiness for operation shall be less than one hour.

Analysis: According to the proper design of the FlexLEU, the HW modules can be exchanged easily. The replacement of a single HW module can be done within less than one hour (MTTR = 1h) assuming that no other faults are to repair. Start-up tests and similar activities necessary for checking readiness for operation are included within the time frame.

Result: Requirement fulfilled

[FlexLEU-SSS-REQ-0433]: FlexLEU - I/O failure behavior, overall operation
[FlexLEU-SSS-SL-REQ-1575]: FlexLEU slices - I/O failure behvaiour, other slice operation

Description: If a failure in an I/O slice input or output occurs while the communication with the slice stays intact, the system shall stay operational with the other inputs and outputs.

Analysis: This is assured by the slice controller concept. The SC of the I/O slice provides communication to the other slices even if a functional outage of the I/O slice occurs. This is also analysed in [RAM_HW_FMEA_FlexLEU].

Result: Requirement fulfilled

[FlexLEU-SSS-REQ-0435]: FlexLEU - I/O slice failure behavior
[FlexLEU-SSS-SL-REQ-1577]: FlexLEU slices - I/O slice failure behaviour

Description: If an I/O slice fails in a way that the communication with the slice is not possible, the FlexLEU shall stay operational with the other reachable I/O slices.

Analysis: This is assured by the logic of the PACS, which keeps the communication to all other reachable slices intact. In case of a bus architecture this refers to all I/O slices to the left of the affected I/O slice and in case of a ring-based architecture this refers to all other I/O slices (see 3.4 and also [RAM_HW_FMEA_FlexLEU]).

Result: Requirement fulfilled

[FlexLEU-SSS-REQ-0436]: FlexLEU - I/O slice communication availability
[FlexLEU-SSS-SL-REQ-1578]: FlexLEU slices - I/O slice communication availability

Description: If a single I/O slice fails the communication to other slices shall stay intact.

Analysis: This is assured by the ring-based architecture in contrast to the plain bus architecture (see 3.4 and also [RAM_HW_FMEA_FlexLEU]).

Result: Requirement fulfilled

7.2.3 MAINTAINABILITY

[FlexLEU-SSS-REQ-0130]: Preventive maintenance

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Description: The system shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0533]: FlexLEU slices - Preventive maintenance – PACS
[FlexLEU-SSS-HW-REQ-0055]: PACS HW - Preventive maintenance

Description: The PACS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_PACS].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0731]: FlexLEU slices - Preventive maintenance – BIS
[FlexLEU-SSS-HW-REQ-0104]: BIS HW - Preventive maintenance

Description: The BIS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_BIS.A] and [ARPA_BIS.B].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0813]: FlexLEU slices - Preventive maintenance – CSS
[FlexLEU-SSS-HW-REQ-0142]: CSS HW - Preventive maintenance

Description: The CSS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_CSS.A] and [ARPA_CSS.B].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0895]: FlexLEU slices - Preventive maintenance – VSS
[FlexLEU-SSS-HW-REQ-0178]: VSS HW - Preventive maintenance

Description: The VSS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_VSS].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-0977]: FlexLEU slices - Preventive maintenance – DIS
[FlexLEU-SSS-HW-REQ-0212]: DIS HW - Preventive maintenance

Description: The DIS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

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Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_DIS].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-1054]: FlexLEU slices - Preventive maintenance – DOS
[FlexLEU-SSS-HW-REQ-0250]: DOS HW - Preventive maintenance

Description: The DOS HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [ARPA_DOS].

Result: Requirement fulfilled

[FlexLEU-SSS-SL-REQ-1175]: FlexLEU slices - Preventive maintenance – PSU
[FlexLEU-SSS-HW-REQ-0320]: PSU HW - Preventive maintenance

Description: The PSU HW shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory lifetime (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU] and also in [DS_PSU_24VAC] and [DS_PSU_24VDC].

Result: Requirement fulfilled

[FlexLEU-SSS-HW-REQ-0661]: Switch HW - Preventive maintenance

Description: The Switch shall not contain electronic components requiring preventive maintenance, inspection or change before the end of the mandatory life time (e.g. battery).

Analysis: This requirement is respected as can be seen for instance in Table 111: HSE Management, Topic: RAMS in [SDD_SYS_FlexLEU]. According to [DS_SWITCH], no fans, batteries or other wear parts are used, hence no preventive maintenance necessary.

Result: Requirement fulfilled

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8. SUMMARY

A dependability analysis of the system FlexLEU has been done with focus on the MTBF requirements for small resp. large reference setups for indoor (GB 40°C) and outdoor (GF 40°C) environment. Basis for these requirements were comparable setups from LEU PV 3.6.

- Due to the usage of new technologies, extended functionality also with focus on improved availability (for instance improvements to better cope with voltage stress (see 6.3.1 and note in 6.5)), improved modularity (slice controller concept) and other aspects, these MTBF requirements could not verbatim fulfilled.
- The MTBF calculation of HW-slices of FlexLEU leads to a deviation by a factor 1.3 to 1.6 from the requirements for the reference setups (see 6.3.1). An analysis of MTBF values of HW components as well as a repair analysis regarding defective components was carried out and documented in [TR_2022-14] (see 6.3 to 6.5).
- An admissible and reasonable approach is to consider *empirical MTBF* values from LEU installations in CH. This leads to an improvement by a factor 1.7 and so to a partial fulfilment of the requirements.

This was agreed by Head of Train Control Product Policy as well as Product Management and other stakeholders (see [TR_2022-14]). Note, this approach was already taken during the introduction of SEC-Signal CH and accepted by the customer SBB.

There are subordinate to the MTBF requirements of reference setups, MTBF requirements to I/O slices which are not fulfilled for precisely the reasons stated above.

Requirements for the master slice PACS and all the other analysed dependability requirements apart from I/O slice MTBF requirements are fulfilled.

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